

Synopsis
On

“LeafDoc”

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Guide

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Chapter: 1

Introduction

Agriculture and home gardening play a vital role in food production, environmental balance, and sustainable living. Plant health monitoring is a crucial aspect of maintaining healthy crops and plants, as diseases and improper care can significantly reduce yield and plant quality. Early identification of plant species and timely detection of diseases are essential for effective treatment and prevention.

In real-world scenarios, many farmers, gardeners, and plant enthusiasts face difficulty in identifying plant species accurately. Most existing plant disease detection systems assume that the plant species is already known, which limits their usability for non-expert users. Additionally, traditional disease diagnosis methods rely on expert knowledge, manual inspection, and laboratory testing, which are time-consuming and not always accessible.

With recent advancements in artificial intelligence, particularly generative vision-language models, image-based plant analysis has become more practical and accessible. These models can analyze plant images and generate meaningful insights without requiring specialized training or large labeled datasets. The proposed project, **LeafDoc**, leverages generative AI to provide an intelligent and user-friendly solution for plant identification, disease diagnosis, and plant care guidance through a single web-based platform.

Objective

The primary objective of the **LeafDoc** project is to develop an AI-assisted web application that identifies plant species from an uploaded image and subsequently detects possible diseases affecting the plant.

The specific objectives of the project are:

- To identify plant species using image-based analysis.
- To detect plant diseases after plant identification.
- To provide treatment suggestions and plant care recommendations.
- To integrate an AI-powered chatbot for resolving plant-related queries.
- To develop a user-friendly and accessible web-based system for non-expert users.

Problem Definition

Despite the availability of various agricultural tools and online resources, users still face several challenges related to plant health management:

- Difficulty in identifying plant species accurately.
- Dependence on prior plant knowledge for disease diagnosis.
- Limited access to agricultural experts in rural and remote areas.
- Lack of integrated systems that provide identification, diagnosis, and treatment guidance.
- Time-consuming and inefficient traditional plant disease detection methods.

The **LeafDoc** project aims to address these challenges by providing a centralized, AI-powered solution that assists users in plant identification, disease detection, and plant care through image-based analysis.

Chapter: 2

Literature Review

Several research studies have focused on plant disease detection using image processing and deep learning techniques. Convolutional neural network (CNN)-based models have shown promising results in identifying plant diseases from leaf images. However, most of these approaches assume prior knowledge of the plant species and focus primarily on accuracy metrics.

Other studies have explored image-based plant identification using machine learning models, but these systems often function independently and do not include disease diagnosis or advisory support. Intelligent agricultural advisory systems have also been proposed, but many rely on sensor data or rule-based approaches rather than visual analysis.

Recent advancements in generative artificial intelligence and vision-language models have enabled intelligent interpretation of images without the need for model training or extensive datasets. However, limited research has focused on integrating plant identification, disease diagnosis, and advisory support into a single system. The **LeafDoc** project addresses this research gap by combining these functionalities using generative AI.

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Proposed Methodology

The **LeafDoc** system follows a structured and sequential methodology to analyze plant health using image-based input. The proposed methodology consists of the following stages:

1. **Image Acquisition:**

The user uploads an image of a plant or leaf through the web-based interface.

2. **Plant Identification:**

The uploaded image is analyzed using a generative vision-language AI model to identify the plant species based on visual characteristics.

3. **Disease Diagnosis:**

After plant identification, the system analyzes the image to detect potential diseases affecting the plant.

4. **Recommendation Generation:**

Based on the detected disease, treatment suggestions and plant care guidelines are generated.

5. **Chatbot Interaction:**

An AI-powered chatbot assists users by answering plant-related questions and providing additional guidance.

The system is implemented as a web-based application using secure API-based integration of generative AI services.

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Expected Outcome

The expected outcomes of the **LeafDoc** project include:

- A functional web-based system for plant identification and disease diagnosis.
- Accurate identification of plant species using image-based analysis.
- Detection of common plant diseases and generation of relevant treatment suggestions.
- An interactive chatbot for plant care and disease-related queries.
- Improved accessibility to plant health information for farmers, gardeners, and students.
- A scalable and extendable prototype suitable for future enhancements.

The system aims to support intelligent plant health monitoring and contribute to the adoption of AI-based solutions in agriculture and home gardening.

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